

The empirical recurrence for OEIS A236005: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A236005 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 70$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A236005 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with the minimum plus the maximum equal to the lower median plus the upper median in every 2×2 subblock.”. It has offset 1 and begins

2960, 41608, 587908, 8623452, 130105204, 2020106076, 32207456576, 526051660468, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:59:39, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 70$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 70$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	172	c_{19}	334763400161956411142	c_{37}	−640822966078335150616136992
c_2	−13926	c_{20}	−308133098277652938320	c_{38}	9065818623388826619726869120
c_3	705525	c_{21}	−3246830708278904207703	c_{39}	−5704142861930295923719099136
c_4	−25066528	c_{22}	16265295543225532493709	c_{40}	−16909853026860695030866564864
c_5	662975817	c_{23}	−21375806099314687960665	c_{41}	22696175664488855306877486592
c_6	−13510606424	c_{24}	−94432937685504184441360	c_{42}	19953394077855703836932555776
c_7	216289010295	c_{25}	476512383015505245808140	c_{43}	−48923415904983150296089210880
c_8	−2741246529325	c_{26}	−517669252616090951491410	c_{44}	−11978435162619616667361689600
c_9	27378190080057	c_{27}	−2309767749417839326914474	c_{45}	73837648958849426648607588352
c_{10}	−210313026542704	c_{28}	8909904281894221527042466	c_{46}	−4077087814594952503878107136
c_{11}	1156058155553612	c_{29}	−4330382019066315204676520	c_{47}	−85065620871800464711066976256
c_{12}	−3399277618939502	c_{30}	−42858207488900119749553508	c_{48}	16606850460774622717610164224
c_{13}	−9320075572640652	c_{31}	99171919941662262207477220	c_{49}	78266041126414117933708738560
c_{14}	180942904538936002	c_{32}	44072960753465744644292792	c_{50}	−17015086910081326285469253632
c_{15}	−1092758162622051228	c_{33}	−500964614726768698133677024	c_{51}	−58513572560965562234098941952
c_{16}	3039921111316593660	c_{34}	503557854776292992740193296	c_{52}	8188900356078006594069069824
c_{17}	4482721612492063934	c_{35}	1214253428811893727954255024	c_{53}	34848569380979279937419608064
c_{18}	−81581193797597789335	c_{36}	−3043025782953280979040185168	c_{54}	358898313543051779533438976

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A236005.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.